

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12386222
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Lin; Hsiao-Lang et al.

Display device

Abstract

A display device includes a backlight module and a display module. The backlight module includes a plurality of light emitting units, a base plate and at least one controller. The light emitting units are disposed on a first surface of the base plate, the controller is disposed on a second surface of the base plate, the second surface is opposite to the first surface, and at least one of the light emitting units is electrically connected to the controller. The display module is disposed on the first surface, and the light emitting units are disposed between the display module and the base plate. The backlight module includes a covering layer, an optical film, and a plurality of pins, the covering layer is disposed on the first surface of the base plate, and the pins are disposed between the covering layer and the optical film.

Inventors: Lin; Hsiao-Lang (Miao-Li County, TW), Tsai; Tsung-Han (Miao-Li County, TW), Chen; Jia-Yuan (Miao-Li County, TW)

Applicant: InnoLux Corporation (Miao-Li County, TW)

Family ID: 67540843

Assignee: InnoLux Corporation (Miao-Li County, TW)

Appl. No.: 17/885566

Filed: August 11, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220382108 A1	Dec. 01, 2022

Related U.S. Application Data

continuation parent-doc US 15893707 20180212 ABANDONED child-doc US 17885566

Publication Classification

Int. Cl.: G02F1/13357 (20060101); G02F1/1335 (20060101)

U.S. Cl.:

CPC G02F1/133608 (20130101); G02F1/133528 (20130101); G02F1/133555 (20130101); G02F1/133603 (20130101); G02F1/133614 (20210101); G02F2201/50 (20130101); G02F2202/36 (20130101)

Field of Classification Search

CPC: G02F (1/133608); G02F (1/133528); G02F (1/133555); G02F (1/133603); G02F (1/133614); G02F (2201/50); G02F (2202/36); G02F (1/133557); G02F (2201/307); G02F (1/133611); G02F (1/133612)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6636286	12/2002	Baek	349/114	G02F 1/133514
2014/0168949	12/2013	Zhao	362/97.3	G02B 6/0021
2016/0033715	12/2015	Lee	257/89	G02B 6/0073
2016/0381317	12/2015	Hosoki	349/65	G02B 6/0068
2017/0045778	12/2016	Cheng	N/A	G02F 1/1337
2018/0080625	12/2017	Yamada	N/A	F21V 7/0083
2018/0180942	12/2017	Peterson	N/A	G02F 1/133603
2018/0188606	12/2017	Lee	N/A	G02F 1/133608
2018/0210282	12/2017	Song	N/A	G02F 1/133514

Primary Examiner: Carruth; Jennifer D.

Assistant Examiner: Dobrowolski; Agnes

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. application Ser. No. 15/893,707, filed on Feb. 12, 2018. The content of the application is incorporated herein by reference.

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

(1) The present disclosure relates to a display device, and more particularly, to a display device including light emitting units having a wide spray angle.

2. Description of the Prior Art

(2) Flat panel display devices, such as liquid crystal display (LCD), inorganic light-emitting diode (LED) display, organic light-emitting diode (OLED) display, and electroluminescent display (ELD), are information viewing technologies for widespread use in a variety of electronic

apparatuses, as the characteristics of thinness, light weight and low dissipation power are widely recognized. As the thickness of the display device continues to be reduced, the distance or spacing for mixing the light emitted from the light-emitting devices in the backlight module is reduced as well. Accordingly, the backlight module may not provide the uniform light to the display module due to the insufficient spacing, and the performance of the display device therefore degrades.

SUMMARY OF THE DISCLOSURE

(3) The present disclosure provides a display device including a backlight module and a display module. The backlight module includes a plurality of light emitting units, a base plate and at least one controller. The plurality of light emitting units are disposed on a first surface of the base plate, the at least one controller is disposed on a second surface of the base plate, the second surface is opposite to the first surface, and at least one of the plurality of light emitting units is electrically connected to the at least one controller. The display module is disposed on the first surface of the base plate, and the plurality of light emitting units are disposed between the display module and the base plate. The backlight module includes a covering layer, an optical film, and a plurality of pins, the covering layer is disposed on the first surface of the base plate, and the plurality of pins are disposed between the covering layer and the optical film.

(4) The present disclosure provides a display device that includes a backlight module and a display module. The backlight module includes a plurality of light emitting units. The display module is disposed on the backlight module, wherein a spray angle of one of the light emitting units is greater than or equal to 135 degrees and less than 180 degrees.

(5) These and other objectives of the present disclosure will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional schematic diagram of a display device according to a first embodiment of the present disclosure.

(2) FIG. 2 is a schematic diagram illustrating a top view of a portion of a backlight module of the display device according to the first embodiment of the present disclosure.

(3) FIG. 3 is a schematic diagram illustrating various backlight modules according to various examples of a second embodiment of the present disclosure.

(4) FIG. 4 is a cross-sectional schematic diagram of a backlight module according to a third embodiment of the present disclosure.

(5) FIG. 5 is a schematic diagram illustrating various display devices according to various examples of a fourth embodiment of the present disclosure.

(6) FIG. 6 is a schematic diagram illustrating various display devices according to various examples of a fifth embodiment of the present disclosure.

(7) FIG. 7 is a schematic diagram illustrating various display devices according to various examples of a sixth embodiment of the present disclosure.

(8) FIG. 8 is a schematic diagram illustrating various display devices according to various examples of a seventh embodiment of the present disclosure.

(9) FIG. 9 is a schematic diagram illustrating various display devices according to various examples of an eighth embodiment of the present disclosure.

DETAILED DESCRIPTION

(10) The present disclosure may be understood by reference to the following detailed description, taken in conjunction with the drawings as described below. It is noted that, for purposes of illustrative clarity and being easily understood by the readers, various drawings of this disclosure

show a portion of the display device, and certain elements in various drawings may not be drawn to scale. In addition, the number and dimension of each device shown in drawings are only illustrative and are not intended to limit the scope of the present disclosure.

(11) Certain terms are used throughout the description and following claims to refer to particular components. As one skilled in the art will understand, electronic equipment manufacturers may refer to a component by different names. This document does not intend to distinguish between components that differ in name but not function. In the following description and in the claims, the terms “include”, “comprise” and “have” are used in an open-ended fashion, and thus should be interpreted to mean “include, but not limited to . . .”.

(12) It will be understood that when an element or layer is referred to as being “on” or “connected to” another element or layer, it may be directly on or directly connected to the other element or layer, or intervening elements or layers may be presented (indirectly). In contrast, when an element is referred to as being “directly on” or “directly connected to” another element or layer, there are no intervening elements or layers presented.

(13) It should be noted that the technical features in different embodiments described in the following may be replaced, recombined, or mixed with one another to constitute another embodiment without departing from the spirit of the present disclosure.

(14) Referring to FIG. 1 and FIG. 2, FIG. 1 is a cross-sectional schematic diagram of a display device according to a first embodiment of the present disclosure, FIG. 2 is a schematic diagram illustrating a top view of a portion of a backlight module of the display device according to the first embodiment of the present disclosure, wherein the cross-sectional structure shown in FIG. 1 corresponds to a line I-I' in FIG. 2. As shown in FIG. 1, a display device **10** includes a backlight module **100** and a display module **200** disposed on an emitting surface of the backlight module **100**. The backlight module **100** can emit light passing through the display module **200**, so that the display module **200** can display an image. The backlight module **100** includes a plurality of light emitting units **102** for generating light L. The light emitting units **102** are inorganic light emitting diodes or organic light emitting diodes. In each light emitting unit **102**, luminance (L_x) of the light L is related to emission angle of the light L, emission angle is an included angle between an emission direction and a normal direction, and each light emitting unit **102** may have maximum L_x at a specific emission angle with respect to the normal direction of an emitting surface of the light emitting unit **102**, for example, light with the maximum L_x (100% maximum L_x) may be emitted from a center of the emitting surface of the light emitting unit **102** along the normal direction, which means the specific emitting angle is 0 degree corresponding to the normal direction, but not limited thereto. In at least one of the light emitting units **102** of this disclosure, an angle between opposite sides of 30% maximum L_x is defined as a spray angle θ which is designed to be used as the light emitted from the backlight module **100**, and the spray angle θ is greater than or equal to 135 degrees and less than 180 degrees. The spray angle is measured in a cross-section view of angle-to-luminance diagram which is a distribution of luminance corresponding to angle. From the origin point of cross-section of the angle-to-luminance diagram, in positive angle side we could find a specific angle corresponding to 30% maximum L_x , in negative angle side we could find another specific angle corresponding to 30% maximum L_x , and the difference (span) between the two specific angles is the spray angle. For example, the light emitting unit **102** with the above-mentioned light distribution may be a chip scale package (CSP) light emitting diode (LED) without relatively large size, the length or width of the chip scale package light emitting diode may be ranged from 0.1 mm to 10 mm, but not limited thereto.

(15) As shown in FIG. 1 and FIG. 2, the backlight module **100** further includes a base plate **104**, a supporting structure **106**, a first optical film **108**, a plurality of controllers **110**, a protecting layer **112**, and a plurality of conductive lines **114**, **116**. The light emitting units **102** are disposed on the base plate **104** and emit light L toward the display module **200**, and the backlight module **100** of this embodiment may be for example a direct type backlight module. The light emitting units **102**

may be mounted directly on the base plate **104**, and the light emitting units **102** and the base plate **104** having such structure may also be regarded as having the so-called chip on board (COB) structure. The light emitting unit **102** may include a light emitting chip **1021** and an encapsulation layer **1022** covering the light emitting chip **1021**. The light emitting chip **1021** in this embodiment can be a blue LED chip, but not limited thereto. In other embodiments, the light emitting chip **1021** may also be a red LED chip, a green LED chip, a white LED chip, an ultra violet (UV) LED chip, or other colors of LED chips. The encapsulation layer **1022** can include materials that can provide protection to the light emitting chip **1021**, the materials may be selected from the group consisting of PI, PC, PE, acrylic, resin, polymer, SiO.sub.x, SiN.sub.x, and polyimide, such materials may have electrical insulation or transparency, but not limited thereto.

(16) In this embodiment, the base plate **104** may include the glass substrate with the conductive lines **114**, **116** and the controllers **110** formed thereon, but not limited thereto. In other embodiments, the base plate **104** may be a printed circuit board (PCB) with plastic or resin based substrate. In this embodiment, the light emitting units **102**, the controllers **110** and the conductive lines **114**, **116** are disposed on a first surface **1041** of the base plate **104**, wherein at least one of the light emitting units **102** is electrically connected to at least one of the controllers **110**. For example, as shown in FIG. 2, each light emitting unit **102** is electrically connected to the controller **110** corresponding thereto by the corresponding conductive line **114**, and the controllers **110** can be electrically connected to each other by the conductive lines **116**. In other embodiment, there are more circuit components to electrically connect elements on the base plate **104** for driving. In this embodiment, the controllers **110** can be disposed in proximity to the light emitting units **102**. For example, the controller **110** may be disposed adjoining to the light emitting units **102** or between two of the light emitting units **102**, as shown in FIG. 1 and FIG. 2. Additionally, the configurations or numbers of the light emitting units **102** and the controllers **110** are not limited to the contents shown in FIG. 1 and FIG. 2. The conductive lines **114** and **116** may be formed of metal or metal oxide. Metal may include Ti, Pt, Au, Ni, Al, Mo, Cu, and the like, as well as alloys of such elements. Metal oxide may include indium tin oxide (ITO) or indium zinc oxide (IZO). The controllers **110** may include an application specific integrated circuit (ASIC) or the like that may be formed directly on the base plate **104** or mechanically mounted to the base plate **104**.

(17) In this embodiment, as shown in FIG. 1 and FIG. 2, the protecting layer **112** is disposed on the base plate **104**. The protecting layer **112** covers at least two light emitting units **102** and at least one controller **110**. A thickness of one of the light emitting units **102** is defined as a first thickness T1, and a thickness Tp of the protecting layer **112** is the same as the first thickness T1. The first thickness Ti is measured from the top of the light emitting unit **102** to the surface of the base plate **104** wherein the light emitting unit **102** is formed on. The thickness Tp of the protecting layer **112** is measured from the top of the protecting layer **112** to the surface of the base plate **104** wherein no other element is formed on. The protecting layer **112** may improve problems such as peeling or scratch issue of the light emitting units **102** or the controllers **110** by external forces, electric leakage, or electrostatic discharge (ESD) damage. Additionally, the coverage of the protecting layer **112** is adjustable. For example, the protecting layer **112** may cover most of the base plate **104**, except the locations of the pins **107** of the supporting structure **106**, but not limited thereto. In another embodiment, the protecting layer **112** may only cover regions of the base plate **104** that the light emitting units **102** and the controllers **110** are disposed thereon or regions of the base plate **104** that stress accumulates therein. In still another embodiment, the backlight module **100** may not include the protecting layer **112**. The protecting layer **112** can include materials that can provide protection to the light emitting units **102** and the controllers **110**, the materials may be selected from the group consisting of PI, PC, PE, acrylic, resin, polymer, SiO.sub.x, SiN.sub.x, and polyimide, such materials may have electrical insulation, transparency, or may be adhesive, but not limited thereto.

(18) The first optical film **108** is disposed on the light emitting units **102**, and the supporting

structure **106** is disposed between the base plate **104** and the first optical film **108**. In this embodiment, the supporting structure **106** includes a plurality of pins **107** for supporting the first optical film **108** and uniforming the spacing between the first optical film **108** and the base plate **104**. In this embodiment, two opposite ends of each pin **107** respectively contact the first optical film **108** and the base plate **104**, but not limited thereto. Since the height of the pin **107** is greater than the first thickness T1 of the light emitting unit **102**, a gap is formed between the first optical film **108** and the light emitting unit **102**. A distance between the first optical film **108** and the light emitting unit **102** is defined as a third distance D3. The third distance D3 can be measured from the light emitting unit **102** to the first optical film **108** in a direction perpendicular to the surface of the first optical film **108**. In addition, a ratio of the third distance D3 to the first thickness T1 ($D3/T1$) is ranged from 1.5 to 75. When the third distance D3 is too small (i.e., the ratio is less than 1.5), the mixing effect of the lights L generated from the light emitting units **102** may be poor, and the light provided by the backlight module **100** may have poor uniformity. When the third distance D3 is too large (i.e., the ratio is greater than 75), the ratio of brightness of the final output light of the backlight module **100** to brightness of light emitted from the light emitting units **102** may be too less, so the light exiting efficiency of the backlight module **100** is reduced.

(19) The first optical film **108** may collimate the light L emitted from the light emitting units **102** and guide the light L to the display module **200**. The first optical film **108** may be a multi-layer structure and may include for example a light guide, a diffusion sheet, a light enhancement sheet, prism sheets, a protection sheet, etc., but not limited thereto. In other embodiments, the first optical film **108** may be a single-layer structure.

(20) The pins **107** may be formed of insulating materials (e.g., PI, PC, PE, acrylic, resin, or polymer), but not limited thereto. In addition, a cross-sectional profile of the pin **107** is a trapezoid, cylinder, or prism, but not limited thereto.

(21) As shown in FIG. 1, the display module **200** includes a first substrate **202**, a second substrate **204**, an active matrix (AM) layer **206**, a liquid crystal layer **208**, a first polarizer **210**, a second polarizer **212**, and a second optical film **214**. The first substrate **202** and the second substrate **204** are disposed opposite to each other, and the liquid crystal layer **208** is interposed between the first substrate **202** and the second substrate **204** to form a liquid crystal cell. The first substrate **202** and the second substrate **204** may include transparent substrates respectively, for example, each transparent substrate may be a rigid substrate such as a glass substrate or a quartz substrate, or a flexible substrate such as a plastic substrate, but not limited thereto. The plastic substrate may include material such as PI, PC, or PET. The active matrix layer **206** is disposed between the first substrate **202** and the liquid crystal layer **208**. For example, plural thin film transistors (TFTs) and other circuit components may be disposed on a surface of the first substrate **202** for controlling the electric field applying to the liquid crystal layer **208**.

(22) In this embodiment, the first polarizer **210** is disposed between the liquid crystal layer **208** and the first substrate **202**, the second polarizer **212** is disposed between the liquid crystal layer **208** and the second substrate **204**, and the first polarizer **210** and the second polarizer **212** may be so-called inner polarizers because they are positioned inside a liquid crystal cell. In conventional, polarizers are attached on outside surfaces of liquid crystal cell. The first polarizer **210** and the second polarizer **212** in this embodiment may be wire grid polarizers (WGP), but not limited thereto.

(23) In addition, a light converting layer **216** includes a plurality of quantum dots (QD) and a blocking portion **2161**. The blocking portion **2161** includes a plurality of apertures **2161R**, a plurality of apertures **2161G**, and a plurality of apertures **2161B**, wherein each aperture can correspond to an area of a pixel (sub-pixel). The blocking portion **2161** may include any material that can block light, such as metal material, black inorganic material, black organic materials or photoresist materials. For example, the material generally used for forming the black matrix may be adopted to form the blocking portion **2161**. The quantum dots in this embodiment include a plurality of first quantum dot portions **2162G** and a plurality of second quantum dot portions

2162R. Each of the first quantum dot portions **2162G** is disposed in an aperture **2161G** (a first pixel **P1**). The quantum dot material(s) or particles in the first quantum dot portions **2162G** are capable of converting the light **L** (i.e. blue light in this embodiment, UV light in other embodiments) emitted from the light emitting units **102** into green light, and therefore the first pixels **P1** are green pixels. Similarly, each of the second quantum dot portions **2162R** is disposed in an aperture **2161R** (a second pixel **P2**), the quantum dot material(s) or particles in the second quantum dot portions **2162R** are capable of converting the light **L** emitted from the light emitting units **102** into red light, and therefore the second pixels **P2** are red pixels. In this embodiment, the apertures **2161B** correspond to a plurality of third pixels **P3**, wherein the third pixels **P3** are blue pixels. Since the light emitting units **102** emits blue light, it is not necessary to additionally dispose blue quantum dot material(s) or particles in the aperture **2161B**, and the blue light can emit through the apertures **2161B** directly, but not limited thereto. In addition, the pixel mentioned herein may be a sub-pixel or a pixel of an image displayed from the display module. Colors of the pixels of this disclosure are not limited to the description above.

(24) In this embodiment, a distance (from center to center, or from edge to edge, a repeatable range) between adjacent two of the light emitting units **102** is defined as a first distance **D1**, and a distance (from center to center, or from edge to edge, a repeatable range) between adjacent two of the pixels is defined as a second distance **D2**, wherein a ratio of the first distance **D1** to the second distance **D2** ($D1/D2$) is ranged from 3 to 1000, and such ratio is defined as LED distance per pixel distance (**LDPPD**) in this disclosure. The first distance **D1** may be calculated from a location in a light emitting unit **102** to the same location in another light emitting unit **102** adjacent thereto, and the second distance **D2** may be calculated from a location in a pixel (e.g., a first pixel **P1**, a second pixel **P2**, or a third pixel **P3**) to the same location in another pixel adjacent thereto. For example, the first distance **D1** may be calculated from a center of a light emitting unit **102** to a center of another light emitting unit **102** adjacent thereto, and the second distance **D2** may be calculated from a center of a pixel (e.g., a first pixel **P1**, a second pixel **P2**, or a third pixel **P3**) to a center of another pixel adjacent thereto, but not limited thereto.

(25) In addition, the widths of the first pixel **P1**, the second pixel **P2**, and the third pixel **P3** may be different, and therefore the distances between adjacent two of the pixels in a group including the first pixels **P1**, the second pixels **P2**, and the third pixels **P3** may also be different. In this situation, the second distance **D2** is defined by the shortest distance between adjacent two of the pixels in the above group of pixels.

(26) The **LDPPD** is a reference for designs such as local dimming or high dynamic range (**HDR**) of the display device **10**. For example, the light emitting unit **102** in this embodiment is **CSP LED**, its size is less than the conventional **LED**, so the number of **CSP LEDs** that can be disposed in a certain area of the base plate **104** may be more comparing to the display device using conventional **LEDs** in the same condition, and the smaller first distance **D1** may be obtained in this embodiment. Therefore, the **LDPPD** of the display device **10** in this embodiment may be less than that of the display device having conventional **LEDs**. For example, the **LDPPD** in this embodiment is ranged from 3 to 200, but not limited thereto. The display device **10** may thus achieve better control of local dimming and have ability to adjust the brightness of a smaller region of pixels.

(27) The shape of each light emitting unit **102** may include shapes such as square, rectangle, pentagon, trapezoid, hexagon, circle, ellipse, rhombus, octagon, etc. In one embodiment, light emitting units **102** may be divided into different groups which may be disposed on different regions of the base plate **104**. The light emitting units **102** in each group may be disposed in the matrix arrangement or in the brick arrangement. For example, the light emitting units **102** in some groups may be disposed in the matrix arrangement and the light emitting units **102** in other groups may be disposed in the brick arrangement. In another embodiment, different groups may have different power consumptions. For example, the power consumption of some of the groups may be 2 watts, the power consumption of the other groups may be 1 watt, and the brightness of the light emitting

area may be different when the power consumptions are different.

(28) The second optical film **214** is disposed on the second substrate **204** to reduce the amount of ambient light penetrating into the display device **10** and prevent the ambient light from reacting with the QD in the light converting layer **216**. The second optical film **214** may include the multi-layer structure, in which several layers having high refractive index or low refractive index may stack together, but not limited thereto. In another embodiment, the second optical film **214** may include a semi-transmittance structure, in which at least one reflection layer, at least one inorganic layer, and at least one metal layer may stack together, and the stacked layers may have openings, but not limited thereto. In other embodiment, the second optical film **214** may be quarter wavelength plate.

(29) The following description will detail the different embodiments of the present disclosure. To simplify the description, identical components in each of the following embodiments are marked with identical symbols. For making it easier to understand the differences between the embodiments, the following description will detail the dissimilarities among different embodiments and the identical features will not be redundantly described.

(30) Referring to FIG. 3, FIG. 3 is a schematic diagram illustrating various backlight modules according to various examples of a second embodiment of the present disclosure. As shown in the part (A) of FIG. 3, the difference between the backlight module **100** in the part (A) of this embodiment and the first embodiment is that the thickness T_p of the protecting layer **112** is greater than the first thickness T_1 of the light emitting unit **102** in the part (A) of this embodiment. In addition, a distance between the first optical film **108** and the protecting layer **112** is defined as a fourth distance D_4 . The fourth distance D_4 can be measured from a top surface of the protecting layer **112** to the bottom surface of the first optical film **108** in a direction perpendicular to the bottom surface of the first optical film **108** or the top surface of the protecting layer **112**. In addition, a ratio of the fourth distance D_4 to the first thickness T_1 (D_4/T_1) is ranged from 1.5 to 75. As shown in the part (B) of FIG. 3, the difference between the backlight module **100** in the part (B) of this embodiment and the first embodiment is that the thickness T_p of the protecting layer **112** is less than the first thickness T_1 of the light emitting unit **102** in the part (B) of this embodiment. As shown in the part (C) and the part (D) of FIG. 3, the difference between the backlight module **100** in the part (C) and the part (A) and the difference between the backlight module **100** in the part (D) and the part (B) in this embodiment are that the supporting structure **106** is disposed on the protecting layer **112** and between the protecting layer **112** and the first optical film **108**. Additionally, two opposite ends of each pin **107** respectively contact the first optical film **108** and the protecting layer **112**, but not limited thereto.

(31) Referring to FIG. 4, FIG. 4 is a cross-sectional schematic diagram of a backlight module according to a third embodiment of the present disclosure. As shown in FIG. 4, the difference between this embodiment and the first embodiment is that the supporting structure **106** includes a plurality of cavities **106C**, and each of the light emitting units **102** is disposed in one of the cavities **106C**. The supporting structure **106** may be formed of insulating materials (e.g., resin), but not limited thereto. A reflective layer **118** is disposed on side surfaces and top surface of the supporting structure **106**, but not limited thereto. The reflective layer **118** may be formed of reflective materials (e.g., metal or multi-layer reflective structure), but not limited thereto. The light L emitted from the light emitting unit **102** can be reflected in the cavity **106C** by the reflective layer **118**, so that the loss of the light L can be reduced and the light efficiency of the backlight module **100** can be improved. In addition, the backlight module **100** in this embodiment may not include the protecting layer **112** in the first embodiment, but not limited thereto. In another embodiment, the protecting layer can be filled in each of the cavities and cover the light emitting units. In still another embodiment, the supporting structure and the protecting layer can together form a flat top surface, and an adhesive layer can be disposed between the flat top surface and the first optical film **108**, such that the backlight module and the display module can be bonded by the adhesive layer.

The adhesive layer may include optically clear adhesive (OCA), but not limited thereto.

(32) Referring to FIG. 5, FIG. 5 is a schematic diagram illustrating various display devices according to various examples of a fourth embodiment of the present disclosure. As shown in the part (A) of FIG. 5, the difference between the backlight module **100** in the part (A) of this embodiment and the first embodiment is that the light emitting units **102** and the controllers **110** are disposed on opposite surfaces of the base plate **104**. For example, the light emitting units **102** are disposed on the first surface **1041** of the base plate **104**, and the controllers **110** are disposed on a second surface **1042** of the base plate **104** opposite to the first surface **1041**. In addition, at least one light emitting unit **102** is electrically connected to at least one controller **110**. For example, as shown in the part (A) of FIG. 5, two light emitting units **102** are respectively electrically connected to one controller **110** by two conductive lines **114** through two via holes **120** penetrating through the base plate **104**. Additionally, in this embodiment, the base plate **104** may be the glass substrate, and through glass vias (TGVs) or through plastic vias (TPVs) may be applied to the base plate **104** to electrically connect the light emitting units **102** and the controller **110**, but not limited thereto.

(33) As shown in the part (B) of FIG. 5, at least one light bar is applied to the backlight module **100**. For example, the light emitting units **102** may be mounted on the base plate **104** and disposed in a strip arrangement to form the light bar. The base plate **104** may be for example PCB that can electrically connect the light emitting units **102** disposed thereon to one or more driving devices.

(34) As shown in the part (C) of FIG. 5, the backlight module **100** further includes an active matrix layer **122** disposed between the light emitting units **102** and the base plate **104**. The active matrix layer **122** includes a semiconductor layer **124**, a dielectric layer **126**, a conductive layer having gate electrodes **128G**, a dielectric layer **130**, a conductive layer having drain electrodes **132** and source electrodes **134**, and an optional dielectric layer **136**, so as to form a plurality of TFTs **1281**, **1282** disposed between the light emitting units **102** and the base plate **104**. The TFTs **1281**, **1282** serve as switch elements for driving the light emitting units **102** in the backlight module **100**. The semiconductor layer **124** is formed of a semiconductor material, such as silicon or metal oxide, but not limited thereto. For example, the semiconductor layer **124** may be amorphous silicon, low temperature poly-silicon, or indium gallium zinc oxide (IGZO). The semiconductor layer **124** includes a source contact **128S**, a drain contact **128D**, and a channel **128C** disposed between the source contact **128S** and the drain contact **128D** in one TFT. Each source electrode **134** is electrically connected to the corresponding source contact **128S** through a via hole in the dielectric layer **130** and the dielectric layer **126**. Each drain electrode **132** is electrically connected to the corresponding drain contact **128D** through another via hole in the dielectric layer **130** and the dielectric layer **126**. The gate electrode **128G** is separated from the channel **128C** by the dielectric layer **126** (e.g., a gate dielectric layer). The gate electrode **128G**, the source electrode **134** and the drain electrode **132** may be formed of conductive materials (such as metal), but not limited thereto. It should be noted that the structure of the TFTs shown in the part (C) of FIG. 5 is merely an example and is not meant to limit the types or structures of the TFTs of the present disclosure, and any other suitable TFT structures may replace the illustrated TFTs. For example, the TFTs **1281**, **1282** are top-gate type TFTs in this embodiment; however, bottom-gate type TFTs may be used as the TFTs **1281**, **1282** in other embodiments.

(35) In addition, at least one of the light emitting units **102** is electrically connected to at least one of the TFTs **1281**, **1282**. For example, the TFTs **1281**, **1282** are respectively electrically connected to the corresponding light emitting units **102** through connecting electrodes **138**. The connecting electrodes **138** may be formed of conductive materials (such as metal), but not limited thereto. In the part (C) of FIG. 5, each of the light emitting units **102** is electrically to one of the TFTs (TFT **1281** or TFT **1282**), but not limited thereto. In other embodiments, each of the light emitting units **102** may be electrically to plural TFTs, or plural light emitting units **102** may be electrically connected to one TFT. In addition, in the part (C) of FIG. 5, the TFT **1282** is at least partially covered or overlapped by the corresponding light emitting unit **102** disposed thereon, and the TFT

1281 is not covered or overlapped by the corresponding light emitting unit **102** disposed thereon, but not limited thereto. In this situation, the backlight module **100** may further include the supporting structures **106** described in the above embodiments, wherein the supporting structure **106** may be disposed on the active matrix layer **122** and cover the TFT **1281**. In addition, the backlight modules **100** in FIG. 5 may further include the protecting layers **112** described in the above embodiments, so as to cover the light emitting units **102**.

(36) Referring to FIG. 6, FIG. 6 is a schematic diagram illustrating various display devices according to various examples of a fifth embodiment of the present disclosure. For clarity, the supporting structure **106**, the controller **110** and the protecting layer **112** of the backlight module **100** are not shown, but not limited thereto. The backlight module **100** may be any one of the backlight modules described in the above embodiments. In the display modules **200** of this embodiment shown in FIG. 6, one of the polarizers **210**, **212** is disposed between the first light converting layer **216** and the liquid crystal layer **208**. As shown in the part (A) of FIG. 6, one difference between the part (A) of FIG. 6 and the first embodiment is that the first polarizer **210** is disposed between the first substrate **202** and the first optical film **108** in the part (A) of FIG. 6. The first polarizer **210** in the part (A) of FIG. 6 may be so-called outer polarizer (conventional), wherein the first polarizer **210** may be formed of plastic materials, but not limited thereto. Another difference between the part (A) of FIG. 6 and the first embodiment is that the liquid crystal layer **208** further includes partitions **2081** to isolate different portions of the liquid crystal molecules. The partitions **2081** can be disposed corresponding to the blocking portion **2161**, and each portion of the liquid crystal molecules can correspond to one of the pixels, but not limited thereto. The partitions **2081** may be formed of the material generally used for forming the black matrix, but not limited thereto. For example, the partitions **2081** may be formed of resin including carbon or titanium pigment.

(37) As shown in the part (B) of FIG. 6, the difference between the part (B) and the part (A) of FIG. 6 is that the liquid crystal layer **208** includes micro cavity structure in the part (B). In the micro cavity structure, a plurality of tunnel-shaped cavities **208C** are disposed in an organic or inorganic insulation layer, wherein liquid crystal capsules may respectively be disposed in each of the tunnel-shaped cavities **208C**, and each liquid crystal capsule may include liquid crystal molecules surrounded by an insulation layer. An alignment layer may also be formed in each tunnel-shaped cavity **208C**, but not limited thereto.

(38) As shown in the part (C) of FIG. 6, the difference between the part (C) and the part (A) of FIG. 6 is that the liquid crystal layer **208** does not include partitions, and the display module **200** includes a first Bragg layer **218** and a second Bragg layer **220** in the part (C). Bragg layer is one kind of transfective layer with half transparent ability and half reflective ability. The first Bragg layer **218** is disposed between the second polarizer **212** and the light converting layer **216**, and the second Bragg layer **220** is disposed between the light converting layer **216** and the second substrate **204**. Both of the first Bragg layer **218** and the second Bragg layer **220** are formed of layers stacked together, wherein numbers of layers stacked in the first Bragg layer **218** and the second Bragg layer **220** are different, and the thickness of the first Bragg layer **218** and the second Bragg layer **220** are different, but not limited thereto. The first Bragg layer **218** can be a full layer that can cover the light converting layer **216** entirely, but not limited thereto. The first Bragg layer **218** can prevent the light converted by the quantum dot materials (i.e., green or red light in this embodiment) from being reflected back to the liquid crystal layer **208**, or being reflected to the adjacent pixels and affecting colors of the adjacent pixels. The second Bragg layer **220** includes a plurality of apertures **220a**, wherein the apertures **220a** may be disposed corresponding to the apertures **2161B** (i.e., the third pixels P3) for allowing the blue light passing through. The apertures **220a** may be filled with a filling layer or a plurality of scattering particles. The second Bragg layer **220** covers the first quantum dot portions **2162G** and the second quantum dot portions **2162R** (i.e., the first pixels P1 and the second pixels P2), and the second Bragg layer **220** can block the residual blue light that is

not converted by quantum dot materials in the first quantum dot portions **2162G** or the second quantum dot portions **2162R** to maintain colors of the first pixels **P1** and the second pixels **P2**. (39) As shown in the part (D) of FIG. 6, one of the differences between the part (D) of FIG. 6 and the first embodiment is that the light converting layer **216** is disposed between the active matrix layer **206** and the first polarizer **210** in the part (D) of FIG. 6, and the structure of the light converting layer **216**, the active matrix layer **206** and the first substrate **202** can be so-called color filter on array (COA) structure. The display module **200** in the part (D) of FIG. 6 further includes a black matrix **222** disposed between the second polarizer **212** and the liquid crystal layer **208**, so as to prevent the color of each pixel from being affected by light corresponding to other pixels. In addition, the display device **10** in the part (D) of FIG. 6 does not include the first optical film **108** and the second optical film **214** in the first embodiment. In the display device **10** in the part (D), the first substrate **202** can be glass substrate that can serve as glass light guide plate (GLGP), so as to guide the light **L** emitted from the light emitting units **102** to the display module **200**. The second polarizer **212** and the black matrix **222** together can reflect or absorb the ambient light, so as to reduce the amount of ambient light penetrating into the display device **10** and reacting with the QD in the light converting layer **216**.

(40) Referring to FIG. 7, FIG. 7 is a schematic diagram illustrating various display devices according to various examples of a sixth embodiment of the present disclosure. For clarity, the supporting structure **106**, the controller **110** and the protecting layer **112** of the backlight module **100** are not shown, but not limited thereto. The backlight module **100** may be any one of the backlight modules described in the above embodiments. In the display modules **200** of this embodiment shown in FIG. 7, one of the polarizers **210**, **212** is disposed between the first light converting layer **216** and the liquid crystal layer **208**. As shown in the part (A) of FIG. 7, the difference between the part (A) of FIG. 7 and the part (A) of FIG. 6 is that the display module **200** in the part (A) of FIG. 7 further includes a third substrate **224**, wherein the second substrate **204** is disposed between the first substrate **202** and the third substrate **224**, the light converting layer **216** is disposed between the second substrate **204** and the third substrate **224**, and the second optical film **214** is disposed on the third substrate **224**. The third substrate **224** may be formed of materials that are the same as the second substrate **204** or the first substrate **202**, but not limited thereto. In addition, another difference is that the liquid crystal layer **208** in the display module **100** in the part (A) of FIG. 7 does not include partitions, but not limited thereto.

(41) As shown in the part (B) of FIG. 7, the difference between the part (B) and the part (A) of FIG. 7 is that the second polarizer **212** is disposed between the first substrate **202** and the light converting layer **216** in the part (B). The second polarizer **212** in the part (B) of FIG. 7 may be so-called outer polarizer (conventional), wherein the second polarizer **212** may be formed of plastic materials for instance, but not limited thereto. As shown in the part (C) of FIG. 7, the difference between the part (C) and the part (A) of FIG. 7 is that the display module **100** in the part (C) does not include the third substrate disposed between the second optical film **214** and the light converting layer **216**. As shown in the part (D) of FIG. 7, the difference between the part (D) and the part (C) of FIG. 7 is that the display module **200** in the part (D) does not include the second optical film and further include a black matrix **222** disposed between the second substrate **204** and the second polarizer **212**.

(42) Referring to FIG. 8, FIG. 8 is a schematic diagram illustrating various display devices according to various examples of a seventh embodiment of the present disclosure. For clarity, the supporting structure **106**, the controller **110** and the protecting layer **112** of the backlight module **100** are not shown, but not limited thereto. The backlight module **100** may be any one of the backlight modules described in the above embodiments. In the display modules **200** of this embodiment shown in FIG. 8, the light converting layer **216** is disposed between one of the polarizers **210**, **212** and the liquid crystal layer **208**. As shown in the part (A) of FIG. 8, one difference between the part (A) of FIG. 8 and the first embodiment is that the light **L** emitted by the

light emitting units **102** is white light, and the light converting layer **216** is color filter in the part (A) of FIG. **8**. The light converting layer **216** may include a plurality of first portions **2163R** for converting white light into red light, a plurality of second portions **2163G** for converting white light into green light, and a plurality of third portions **2163B** for converting white light into blue light, but not limited thereto. The second portions **2163G** may correspond to the first pixels **P1**, the first portions **2163R** may correspond to the second pixels **P2**, and the third portions **2163B** may correspond to the third pixels **P3**. Another difference between the part (A) of FIG. **8** and the first embodiment is that the second polarizer **212** is disposed on the second substrate **204**, and the first polarizer **210** is disposed under the first substrate **202** in the part (A) of FIG. **8**. The second substrate **204** is disposed between the second polarizer **212** and the light converting layer **216**, the first substrate **202** is disposed between the active matrix layer **206** and the first polarizer **210**, the first polarizer **210** and the second polarizer **212** may be so-called outer polarizers, and the first polarizer **210** and the second polarizer **212** may be formed of plastic materials, but not limited thereto. In addition, the display module **200** in the part (A) of FIG. **8** does not include the second optical film.

(43) As shown in the part (B) of FIG. **8**, one of the differences between the part (B) and the part (A) of FIG. **8** is that the light converting layer **216** is disposed between the active matrix layer **206** and the liquid crystal layer **208** in the part (B), and the structure of the light converting layer **216**, the active matrix layer **206** and the first substrate **202** can be so-called color filter on array (COA) structure. The display module **200** in the part (B) further includes the black matrix **222** disposed between the second substrate **204** and the liquid crystal layer **208**, so as to prevent the color of each pixel from being affected by light corresponding to other pixels.

(44) As shown in the part (C) of FIG. **8**, one of the differences between the part (C) and the part (A) of FIG. **8** is that each of the light emitting units **102** further includes another light converting layer **1022'** respectively in the part (C). The light converting layer **1022'** can serve as an encapsulation layer of each light emitting unit **102**, but not limited thereto. The light converting layer **1022'** includes a plurality of quantum dots. For example, blue light emitted by the light emitting chips **1021** can be converted into white light by the light converting layer **1022'**, and therefore the light **L** emitted by the light emitting units **102** is white light, but not limited thereto. In addition, the backlight module **100** in the part (C) does not include the first optical film, but not limited thereto. As shown in the part (D) of FIG. **8**, the difference between the part (D) and the part (C) of FIG. **8** is that the backlight module **100** in the part (D) further includes another light converting layer **140**. The light converting layer **140** in the part (D) is a full layer disposed on the light emitting units **102** and covers the light emitting units **102**. In addition, the light **L** emitted by the light emitting units **102** is blue light, and the light **L** can be convert into white light after passing through the light converting layer **140**.

(45) Referring to FIG. **9**, FIG. **9** is a schematic diagram illustrating various display devices according to various examples of an eighth embodiment of the present disclosure. For clarity, the supporting structure **106**, the controller **110** and the protecting layer **112** of the backlight module **100** are not shown, but not limited thereto. The backlight module **100** may be any one of the backlight modules described in the above embodiments. In the display modules **200** of this embodiment shown in FIG. **9**, the display device **10** further includes a transfective layer **226**, wherein the light converting layer **216** is disposed between the transfective layer **226** and the light emitting units **102**. The transfective layer **226** may be a Bragg layer, or a half transparent metal film. As shown in the part (A) of FIG. **9**, one difference between the part (A) of FIG. **9** and the part (A) of FIG. **6** is that the display module **200** in the part (A) of FIG. **9** further includes the transfective layer **226** disposed between the second substrate **204** and the light converting layer **216**. The transfective layer **226** covers the first pixels **P1** and the second pixels **P2** and does not cover the third pixels **P3**. The transfective layer **226** can absorb the residual blue light that is not converted by the first quantum dot portion **2162G** or the second quantum dot portion **2162R** of the

light converting layer **216** and allow light generated from the first quantum dot portion **2162G** and the second quantum dot portion **2162R** passing through, so as to maintain colors of the first pixels **P1** and the second pixels **P2**. The transfective layer **226** is formed of titanium dioxide (TiO₂), but not limited thereto. In addition, the display module **200** in the part (A) of FIG. **9** does not include the second optical film, and the liquid crystal layer **208** does not include partitions, but not limited thereto.

(46) As shown in the part (B) of FIG. **9**, the difference between the part (B) and the part (A) of FIG. **9** is that the second substrate **204** is disposed between the transfective layer **226** and the light converting layer **216**. As shown in the part (C) of FIG. **9**, the difference between the part (C) of FIG. **9** and the part (D) of FIG. **6** is that the display module **200** in the part (C) of FIG. **9** further includes the transfective layer **226** disposed between the liquid crystal layer **208** and the first polarizer **210**. As shown in the part (D) of FIG. **9**, the difference between the part (D) and the part (C) of FIG. **9** is that the transfective layer **226** is disposed between the first polarizer **210** and the light converting layer **216**.

(47) To summarize the above descriptions, in at least one of the light emitting units in the backlight module, luminance of the light emitted from the light emitting unit greater than or equal to 30% maximum Lx that is within the spray angle is designed as the light emitted from the backlight module, and the spray angle is greater than or equal to 135 degrees and less than 180 degrees. Since the spray angle of the light emitted from the light emitting unit is greater than the beam angle of the light emitting unit in the conventional display device, more light from the light emitting unit can be used as the light emitted from the backlight module in this disclosure, and the light from the backlight module may be more uniform when the thickness of the backlight module is reduced. The protecting layer can cover the light emitting units or the controllers in the backlight module. The protecting layer can improve problems such as peeling or scratch issue of the light emitting units or the controllers by external forces, electric leakage, or electrostatic discharge (ESD) damage. A ratio of the third distance (distance between the first optical film and the light emitting unit) to the first thickness (thickness of the light emitting unit) is ranged from 1.5 to 75 to obtain better light exiting efficiency of the backlight module. In addition, LED distance per pixel distance (LDPPD) in this disclosure is ranged from 3 to 1000 to achieve better control of local dimming and have ability to adjust the brightness of a smaller region of pixels.

(48) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the disclosure. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A display device, comprising: a backlight module comprising a plurality of light emitting units, a base plate and at least one controller, wherein the plurality of light emitting units are disposed on a first surface of the base plate, the at least one controller is disposed on a second surface of the base plate, the second surface is opposite to the first surface, and at least one of the plurality of light emitting units is electrically connected to the at least one controller; and a display module disposed on the first surface of the base plate, wherein the plurality of light emitting units are disposed between the display module and the base plate, wherein the backlight module comprises a covering layer, an optical film, and a plurality of pins, and the covering layer is disposed on the first surface of the base plate, wherein the optical film is disposed on the plurality of pins, wherein the covering layer comprises an opening, and at least one of the plurality of pins is disposed in the opening, wherein the at least one of the plurality of pins comprises a flat surface, and the flat surface is contacted with the optical film.

2. The display device of claim 1, wherein the at least one of the plurality of light emitting units

comprises a side surface and a top surface, the covering layer covers a portion of the side surface of the at least one of the plurality of light emitting units, and the covering layer exposes the top surface of the at least one of the plurality of light emitting units.

3. The display device of claim 1, wherein a thickness of the covering layer is less than a thickness of the at least one of the plurality of light emitting units.

4. The display device of claim 1, wherein the covering layer comprises a plurality of holes, and the plurality of light emitting units are disposed in the plurality of holes respectively.

5. The display device of claim 1, wherein the at least one of the plurality of light emitting units comprises a side surface and a top surface, and the covering layer covers the side surface and the top surface of the at least one of the plurality of light emitting units.

6. The display device of claim 1, wherein the at least one of the plurality of light emitting units is electrically connected to the at least one controller by a conductive line through a via hole penetrating through the base plate.

7. The display device of claim 1, wherein the backlight module further comprises a reflective layer disposed on a side surface and a top surface of the at least one of the plurality of pins.

8. The display device of claim 7, wherein a cavity is disposed between two adjacent pins of the plurality of pins, and the at least one of the plurality of light emitting units is disposed in the cavity.

9. The display device of claim 1, wherein the display module comprises a plurality of pixels, a distance between adjacent two of the plurality of light emitting units is defined as a first distance, and a distance between adjacent two of the plurality of pixels is defined as a second distance, wherein a ratio of the first distance to the second distance ranges from 3 to 1000.

10. The display device of claim 1, wherein a distance between the optical film and one of the plurality of light emitting units is defined as a third distance, and a thickness of the one of the plurality of light emitting units is defined as a first thickness, wherein a ratio of the third distance to the first thickness ranges from 1.5 to 75.

11. The display device of claim 1, wherein a distance between the optical film and the covering layer is defined as a fourth distance, and a thickness of one of the plurality of light emitting units is defined as a first thickness, wherein a ratio of the fourth distance to the first thickness ranges from 1.5 to 75.

12. The display device of claim 1, wherein the backlight module comprises a plurality of thin film transistors, the plurality of thin film transistors are disposed between the plurality of light emitting units and the base plate, and the at least one of the plurality of light emitting units is electrically connected to at least one of the plurality of thin film transistors.

13. The display device of claim 1, wherein the display module comprises: a first light converting layer disposed on the plurality of light emitting units; a first polarizer disposed on the backlight module; a second polarizer disposed on the first polarizer; and a liquid crystal layer disposed between the first polarizer and the second polarizer.

14. The display device of claim 13, wherein one of the first polarizer and the second polarizer is disposed between the first light converting layer and the liquid crystal layer.

15. The display device of claim 13, wherein the first light converting layer is disposed between the liquid crystal layer and the first polarizer.

16. The display device of claim 13, wherein the display device comprises a black matrix, and the liquid crystal layer is disposed between the first light converting layer and the black matrix.

17. The display device of claim 13, wherein the at least one of the plurality of light emitting units comprises a second light converting layer, and the second light converting layer comprises a plurality of quantum dots.

18. The display device of claim 13, wherein the display device comprises a transfective layer, and the first light converting layer is disposed between the transfective layer and the plurality of light emitting units.

19. The display device of claim 18, wherein the display module comprises a pixel and another pixel

adjacent to the pixel, and the transflective layer is disposed on the pixel and is not disposed on the another pixel.

20. The display device of claim 1, wherein the base plate is a printed circuit board.

21. The display device of claim 20, wherein the plurality of light emitting units has a light emitting direction, the display module has a light output direction, and the light emitting direction and the light output direction are parallel.

22. The display device of claim 20, wherein the optical film is disposed between the plurality of pins and the display module.
